IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

TRAFFIC THEORY & QUEUEING SYSTEMS

Tuesday 2 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D.P. Mandic

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Special information for students

1. Erlang Loss formula recursive evaluation:

$$E_N(\rho) = \frac{\rho E_{N-1}(\rho)}{N + \rho E_{N-1}(\rho)}$$
$$E_0(\rho) = 1$$

2. Engset Loss formula recursive evaluation (for a fixed M and $p = \alpha/1+\alpha$):

$$e_N = \frac{(M - N + 1)\alpha e_{N-1}}{N + (M - N + 1)\alpha e_{N-1}}$$
$$e_0 = 1$$
$$\alpha = \lambda/\mu$$

3. Traffic capacity on basis of Erlang B formula (next page).

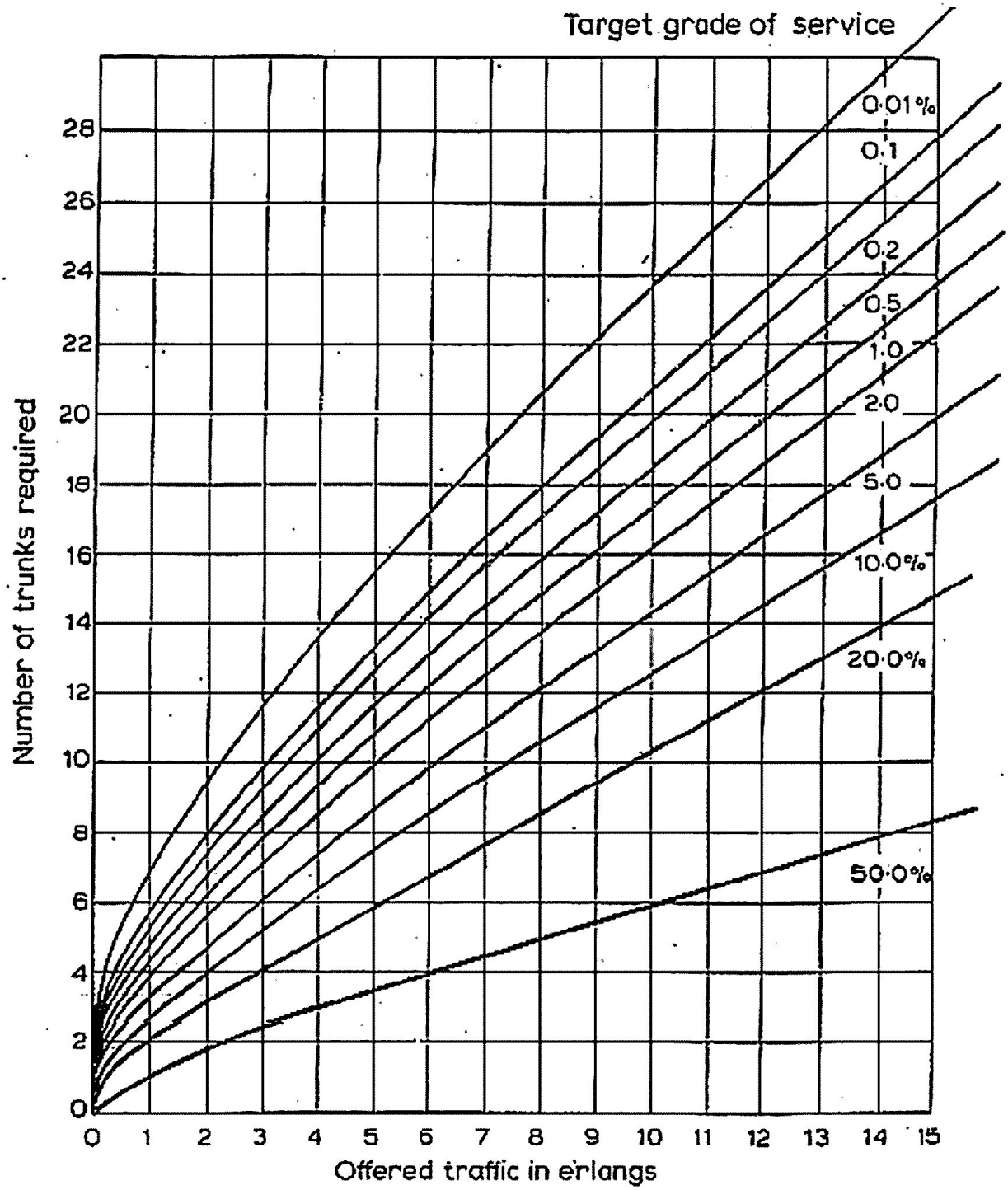
4. Expected residual time

$$E[R] = \frac{1}{2}\lambda E[S^2] = \frac{1}{2}\sum_{k=1}^{m}\lambda_k E[S_k^2]$$

5. Mean and variance of the Kosten overflow model (infinite overflow link $N = \infty$).

$$m = E[N_t] = \rho E_M(\rho)$$

$$v = Var(N_t) = m\left(1 - m + \frac{\rho}{M+m+1-\rho}\right)$$



Traffic capacity on basis of Erlang B. formula.

The Questions

1.

a)

i) Consider an arrival process that has the following characteristics:

- The mean arrival rate is λ.

- The probability of one event in the interval Δt is $\lambda \Delta t$.

- The probability of overlapping arrivals is negligible.

- The probability of events occurring in two nonoverlapping intervals are independent.

Derive the probability that i arrivals occur in an interval Δt as $\Delta t \to 0$. [4]

ii) Introduce and discuss the relationship between a Poisson process and an exponentially distributed random variable, both characterised by the same parameter μ. [3]

iii) For a CTMC model, discuss one condition known to you that will allow you to solve the model using Local Balance Equations (LBE). Use examples to answer the question. [3]

iv) Calls arrive to a message centre at a rate of 30 calls per minute. The mean call duration is 30 seconds. Calculate the loss probability if the centre is endowed with 24 channels. [3]

b) A complaint centre has been split-up into two offices on different floors of a building. They share a 2-circuit link to connect to the network.

The mean arrival rate of one office that has 200 telephones is 0.3 calls per hour. The other office has 10 telephones, and the mean arrival rate is of 6 calls per hour. The call holding time is the same and is 2 minutes.

i) Derive a suitable birth/death model for the traffic offered to this link. State clearly all assumptions made. [4]

ii) For the link which has 2 circuits derive the steady state distribution. [3]

2.

a) M independent ON-OFF sources offer traffic to an N-channel communication link. Each source behaves independently. Assume that p is the probability of one source being busy.

i) State and show under which conditions the probability distribution of the number of busy channels is a truncated binomial probability with parameters M and p. [5]

ii) Given $p = 0.01$, $M = 5$ and $N = 2$, calculate the probability that a request is lost. [5]

b) For the system shown in Fig. 2.1, external arrivals are Poisson distributed with parameter λ and service time is exponentially distributed, with parameter μ.

Assume that the system has 1 server, and that the probability that one customer will require further service is a known probability p.

i) Derive the distribution of customers that are being delayed. [6]

ii) Derive the expected queue length for delay arrivals. [4]

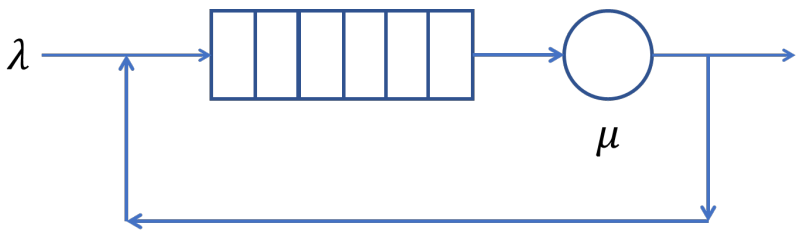


Figure 2.1:

3.

a) For a large N ON-OFF traffic sources multiplexer, it has been suggested that the probability of loss could be approximated by the following expression

$$\varepsilon = \frac{1}{m}\int_{J_0}^{\infty} \frac{e^{-(x-m)^2/2\sigma^2}}{\sqrt{2\pi\sigma^2}}\,dx$$

The integral function ε relates to a random variable x, and to an underlying N ON-OFF traffic sources multiplexer.

i) Discuss the meaning of the integral function ε. Explain how the integral function ε relates to an underlying N ON-OFF traffic sources multiplexer. [5]

ii) Discuss the meaning of x, m and J_o. [4]

b)

Messages, which are composed of one or more packets, arrive to a communication link, at an average rate of 1000 messages per second. The communication link has a capacity to transmit at a rate of 2000 Mbits/s.

Assuming that the message length is uniformly distributed between on the range (1 to 5) packets; and that the packet length is 20 bits, derive:

i) The expected service time $E[s]$. [4]

ii) The mean square service time $E[S^2]$. [4]

iii) The mean message waiting time of the system. [3]

4.

a) Consider the K channel transmission link with a FIFO queue discipline. Two different arriving streams contend for the resources. All arrivals have the same service requirement that is exponentially distributed with mean service rate μ.

Arrivals from stream one are allowed to wait if all K servers are busy.

Arrivals from stream two are loss if all servers are busy.

Assuming that the rate of arrival of stream one is λ_1 and that of stream two is λ_2:

i) Define the state space of the system. [2]

ii) Derive the Markov chain representation of the system. Explicitly identify and include all transition rates. [4]

b) You are asked to analyse an $N(=3)$ redundant computer system. In the reminder of the question assume that the system is operational if at least one computer is running.

Using all the identified states of the system answer the following questions:

i) Identify all transitions that will make this system a non-repairable system. [3]

ii) Identify all possible repair/restoration strategies that you could add to the system described in i). [5]

iii) Modify the system represented in ii) to consider a coverage factor, $c = 0.999$. Explicitly show all modified transition rates. [3]

Now, you are required to augment the model with a reward structure. Consider the system described in ii) and the reward structure $R = [r_0, r_1, r_2, r_3]$. Answer the following question:

iv) Discuss and derive a suitable reward structure that will enable you to obtain the system availability. [3]